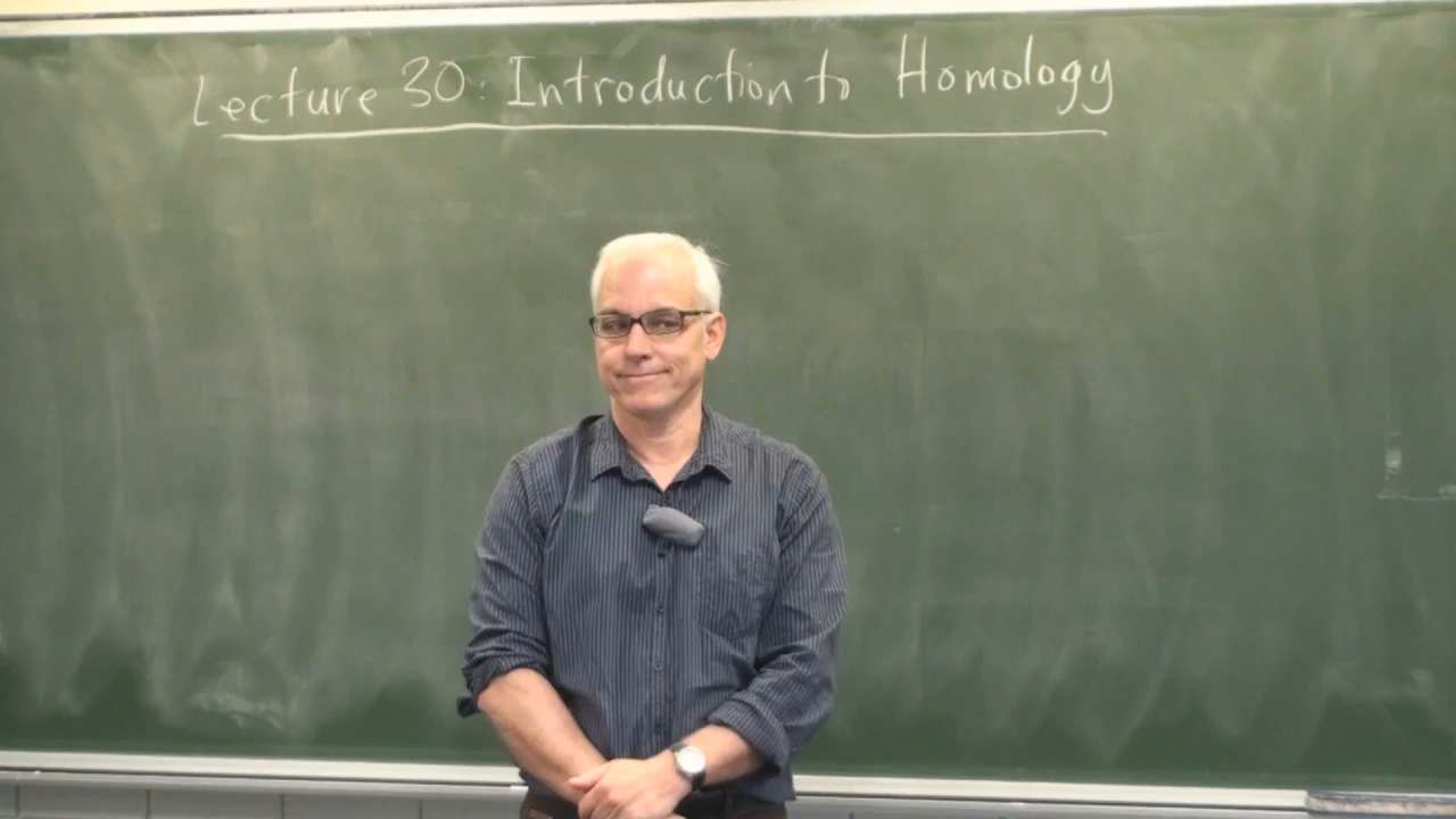
# Algebraic Topology: a beginner's course

This lecture:

# AlgTop30: An introduction to homology


Lecture 30: Introduction to Homology

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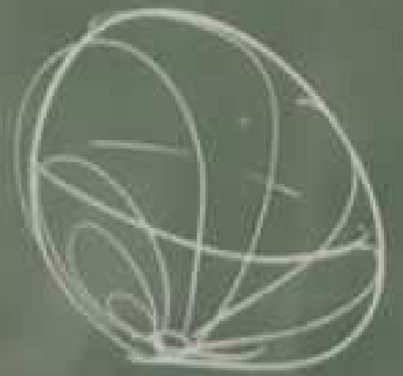
…ere are higher dimensional analogs

homotopy groups $\pi_n(X)$, $n \geq 2$

$\pi_2$: (loops of loops)



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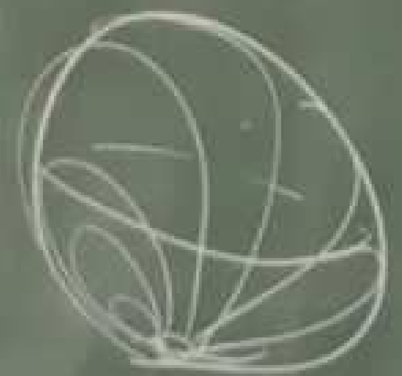
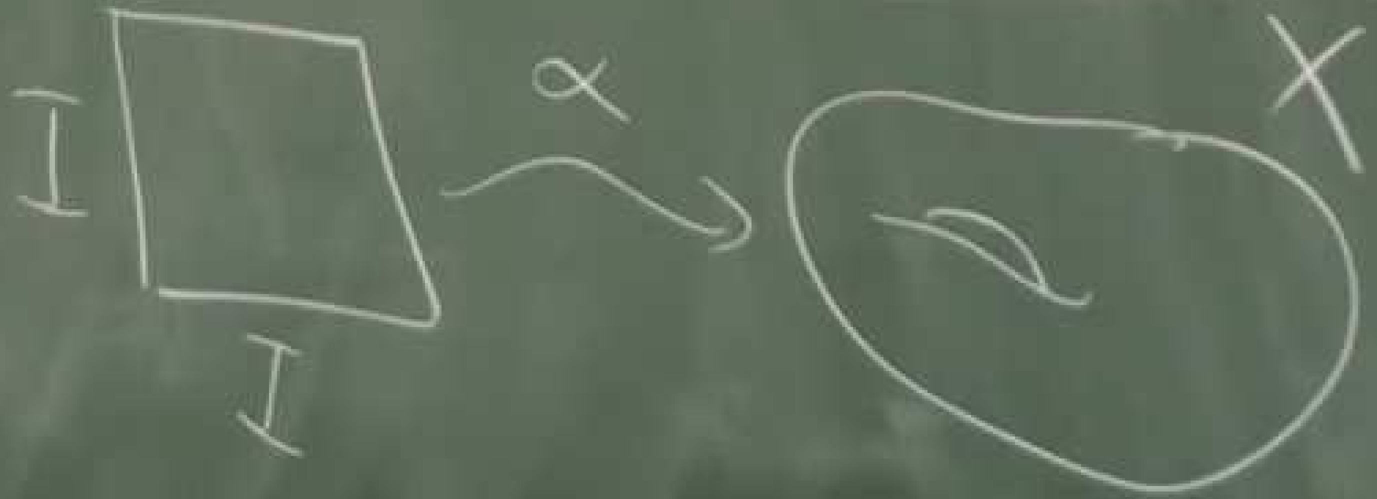




Homology is a <u>commutative</u> alternative to homotopy.

<u>$X$ top space</u> $\Longrightarrow$ $H_n(X)$ homology groups

$n = 0, 1, 2, \ldots$ <u>all</u> commutative.

$\boxed{H_n(X) \approx}$ measures # of $n$-dim holes in $X$.

We will introduce the ideas via examples.

$X$ graph

vertices $x, y, z$

edges $a, b, c, d$.


loop $abc$ starts & ends at $x$

$bca$ " " " $y$

$cab$ " " " $z$

Commutativity $\Rightarrow$ doesn't matter where we start

Also we prefer to write operations as addition,

so loop is represented by $a+b+c = b+c+a = a+c+b$ etc

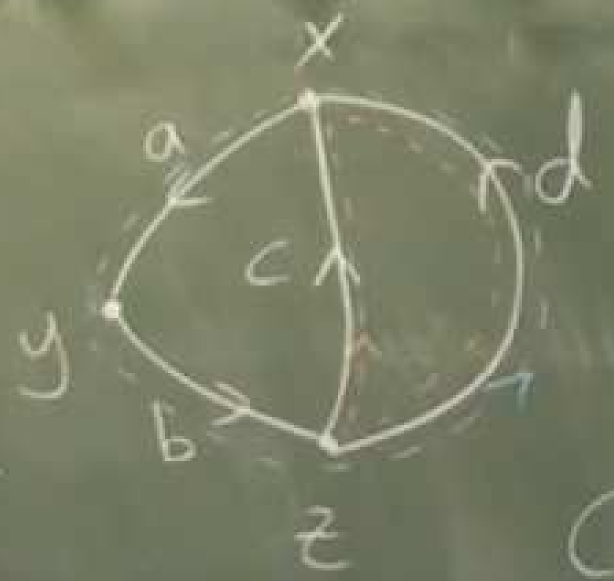
We call this expression a <u>cycle</u> because it is a closed loop geom

A

X graph

vertices $x, y, z$

edges $a, b, c, d$.




loop $abc$ starts & ends at $x$

$bca$ " $y$

$cab$ " $z$

Commutativity $\Rightarrow$ doesn't matter where we start

Algebraic relation:

$$a+b+c = (a+b+d) + (c-d)$$

Also we prefer to write operations as addition

so loop is represented by $\boxed{a+b+c} = b+c+a = a+c$

We call this expression a cycle because it is a closed loop

Another cycle: $\boxed{c-d}$. Another is $\boxed{a+b+d}$.

free abelian group
on vertices $x, y, z$

free abelian group
on (directed edges)
$a, b, c, d$

of $C_0$ are integral linear
$x, y, z$ such as $2x-3y+4z$

nese <u>0-dim</u> <u>chains</u>

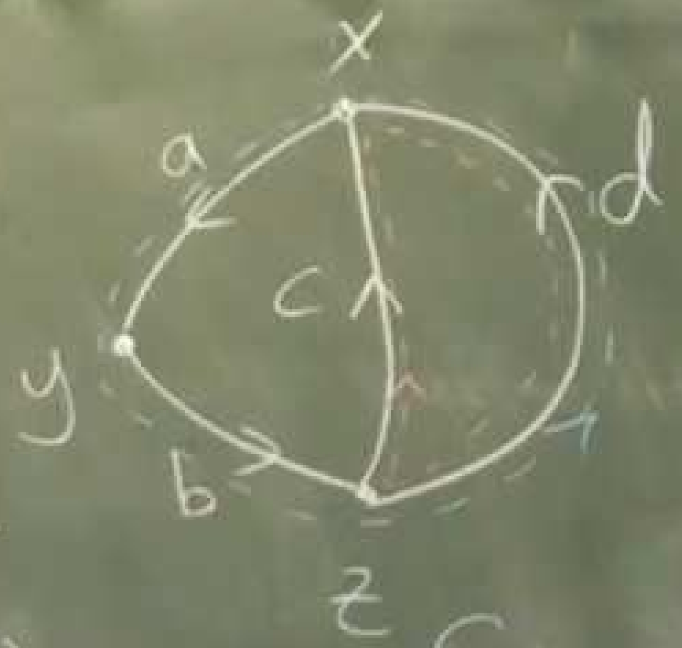
Elements of $C_1$ are integral linear combs
of $a, b, c, d$, such as $a+b+d$, or $2a+b-5c+4d$.

We call these <u>1-dim chains</u>.

Q. What do we mean by a <u>cycle</u>
algebraically?

i.e. what's special about
$a+b+d$? Not shared by
$2a+b-5c+4d$?

$$\partial(\alpha a+\beta b+\gamma c+\delta d)$$
$$=\alpha\,\partial(a)+\beta\,\partial(b)+\gamma\,\partial(c)+\delta\,\partial(d)$$
$$=\alpha(y-x)+\beta(z-y)+\gamma(x-z)+\delta(x-z)\quad(*)$$
$$=(-\alpha+\gamma+\delta)x+(\alpha-\beta)y+(\beta-\gamma-\delta)z$$




Note $\partial(a+b+c)=(y-x)+(z-y)+(x-z)$
$=0$

A. Relationship between e

Each edge has a bound

$\partial(a)=y-x$
$\partial(b)=z-y$
$\partial(c)=x-z$
$\partial(d)=x-z$

$\boxed{\partial(\text{edge}}$

$\partial: C_1\longrightarrow C_0$ is the
which extends $(*)$

in $t$ in $C_1$ is a cycle $\iff \partial(t) = 0$.

$\gamma c + \delta d$ is cycle $\iff$

$$\boxed{\begin{aligned} -\alpha + \gamma + \delta &= 0 \\ \alpha - \beta &= 0 \\ \beta - \gamma - \delta &= 0 \end{aligned}}$$

sol'tns? $\boxed{A\vec{x} = \vec{0}}$

$$\begin{pmatrix} 0 & 1 & 1 \\ -1 & 0 & 0 \\ 1 & -1 & -1 \end{pmatrix}$$

$$\sim \begin{pmatrix} 1 & -1 & 0 & 0 \\ -1 & 0 & 1 & 1 \\ 0 & 1 & -1 & -1 \end{pmatrix} \sim \begin{pmatrix} 1 & -1 & 0 & 0 \\ 0 & -1 & 1 & 1 \\ 0 & 1 & -1 & -1 \end{pmatrix} \sim \begin{pmatrix} 1 & -1 & 0 & 0 \\ 0 & 1 & -1 & -1 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

$$\sim \begin{array}{cccc} \alpha & \beta & \gamma & \delta \\ \hline \end{array}$$
$$\begin{pmatrix} 1 & 0 & -1 & -1 \\ 0 & 1 & -1 & -1 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$
$$\begin{array}{cccc} & & r & s \end{array}$$

$$\boxed{\begin{aligned} \alpha &= r + s \\ \beta &= r + s \\ \gamma &= r \\ \delta &= s \end{aligned}}$$

$$\begin{pmatrix} \alpha \\ \beta \\ \gamma \\ \delta \end{pmatrix} = r \begin{pmatrix} 1 \\ 1 \\ 1 \\ 0 \end{pmatrix} + s \begin{pmatrix} 1 \\ 1 \\ 0 \\ 1 \end{pmatrix}$$

basis for nullspace$(A)$

$\boxed{a + b + c}$

$\boxed{a + b + d}$